

Bernoulli's distribution.

1) A machine produces items and each item is defective with probability 0.1 and non-defective with probability 0.9

a) Define Bernoulli's r.v X for item is defective

b) Find pmf, expectation and variance.

Solⁿ:- a) Bernoulli's random variable

$$X(\text{defective item}) = 1$$

$$X(\text{non-defective}) = 0$$

$$b) \text{ pmf} = p(x) = (0.1)^x (0.9)^{1-x} \quad x=0,1$$

$$E[X] = p = 0.1$$

$$\begin{aligned} \text{Var}(X) &= p(1-p) = 0.1(1-0.1) = 0.1 \times 0.9 \\ &= 0.09 \end{aligned}$$

2) A student guesses the answer to a true or false question.

a) Define Bernoulli's r.v X for answer is correct.

b) Find pmf, expectation and variance.

Solⁿ: Random variable is

$$\begin{aligned} \text{a) } X(\text{correct answer}) &= 1 \\ X(\text{wrong answer}) &= 0 \end{aligned}$$

b) Here the probability that the answer is correct is 0.5

$$P(X=1) = 0.5$$

$$P(X=0) = 1 - 0.5 = 0.5$$

$$\text{pmf} = P(X) = (0.5)^x (0.5)^{1-x} \quad x=0,1$$

$$E(X) = p = 0.5$$

$$\text{Var}(X) = p(1-p) = 0.5 \times 0.5 = 0.25$$

Binomial distribution

1) Disks produced by a company will be defective with probability 0.01. The company sells the disks in a packet of 10 offers money back guarantee if more than one defective disk found in a packet. What is the proportion of packets returned? If someone buys 3 packets, what is the probability that exactly one of them should be replaced?

Given:-

- Probability a disk is defective = $p = 0.01$
- Number of disks per packet = $n = 10$
- A packet is returned if more than one defective disk is found.

∴ X is binomial ($n = 10, p = 0.01$)

We have to find

$$\begin{aligned}P(\text{Packet returned}) &= P(X > 1) \\&= 1 - P(X \leq 1) \\&= 1 - [P(X = 0) + P(X = 1)]\end{aligned}$$

$$P(X = x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$$\begin{aligned}P(X = 0) &= \binom{10}{0} (0.01)^0 (0.99)^{10} = (0.99)^{10} \\&= 0.90438\end{aligned}$$

$$P(X = 1) = \binom{10}{1} (0.01)^1 (0.99)^9 = 0.09135$$

$$\begin{aligned}\therefore P(X \leq 1) &= 0.90438 + 0.09135 \\&= 0.99573\end{aligned}$$

$$\therefore P(\text{Packet returned}) = 0.00427$$

(b) Probability that exactly one of three packets should be replaced.

Let the event "packet returned" = success
 $= p_n = 0.00427$

Now Y is binomial ($n=3$, $p=0.00427$)

We want $P(Y=1) = \binom{3}{1} p_n (1-p_n)^2$

$$P(Y=1) = 3 (0.00427) (1-0.00427)^2 \\ = 0.0127$$

2) Numbers are selected at random one at a time from the 2-digit numbers 00, 01, 02, ..., 99, with replacement. An event A occurs if and only if the product of the 2 digits of a selected number is 18. If 4 numbers are selected, find the probability that the event A occurs at least 3 times.

Solⁿ: Digits are from 0 to 9. We need ordered digit pairs (a, b) with $a \times b = 18$. The valid ordered pairs are $(2, 9), (9, 2), (3, 6), (6, 3)$. Therefore 4 favourable digits out of 100.

So $p = P(A) = \frac{4}{100} = \frac{1}{25}$

$\therefore X \sim \text{Binomial} (n=4, p = \frac{1}{25})$.

We want to find $P(X \geq 3)$

$$\begin{aligned}\therefore P(X \geq 3) &= P(X=3) + P(X=4) \\&= \binom{4}{3} p^3 (1-p)^{4-3} + \binom{4}{4} p^4 (1-p)^{4-4} \\&= 4 p^3 (1-p) + p^4 \\&= 4 \left(\frac{1}{25} \right)^3 \left(1 - \frac{1}{25} \right) + \left(\frac{1}{25} \right)^4 \\&= 4 \left(\frac{1}{25^3} \right) \left(\frac{24}{25} \right) + \frac{1}{(25)^4} \\&= \frac{97}{(25)^4} = \frac{97}{390625}\end{aligned}$$

3. Suppose we roll a fair six-sided die 3 times. What is the probability of getting exactly 2 sixes?

Solⁿ:- Let X be no of sixes obtained in 3 rolls.

Number of trials $= n = 3$

Probability of success (getting a six) $= p = \frac{1}{6}$

Then $X \sim \text{Binomial}(n=3, p=1/6)$

We have to find $P(X=3)$

$$\begin{aligned} P(X=3) &= \binom{3}{3} \left(\frac{1}{6}\right)^3 \left(1 - \frac{1}{6}\right)^{3-3} \\ &= \frac{1}{6^3} \left(\frac{5}{6}\right)^0 = \frac{1}{6^3} = \frac{1}{216} \end{aligned}$$

Normal distribution

Probability density function

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

$\sigma \rightarrow$ standard deviation

$\mu \rightarrow$ mean

Properties

1) If $X_1, X_2, \dots, X_n$ are independent R.V that are normally distributed, then sum S is also normally distributed.

$$S = X_1 + X_2 + \dots + X_n$$

$$E(S) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$V(S) = V(X_1) + V(X_2) + \dots + V(X_n)$$

2) If $X_1, X_2, \dots, X_n$ are independent R.V that are normally distributed, then the R.V. Q is defined as $Q = a_1 X_1 + a_2 X_2 + \dots + a_n X_n + b$ will also be normally distributed with

$$E(Q) = a_1 E(x_1) + a_2 E(x_2) + \dots + a_n E(x_n) + b$$

$$V(Q) = a_1^2 V(x_1) + a_2^2 V(x_2) + \dots + a_n^2 V(x_n)$$

Problems:-

1. The weight of a module used in a spacecraft is to be closely controlled. Since the module uses a bolt-nut-washer assembling in numerous places, a study was conducted to find the distribution of weights of these parts. It was found that the three weights, in grams, are normally distributed with the following means and variance.

	Means	Variances
Bolt	312.8	2.67
Nut	53.2	0.85
Washer	17.5	0.21

Find the distribution of the weight of the assembly. Find mean, variance and standard deviation.

Solⁿ:- The weight of the assembly is the sum of the weights of the 3 component parts, which are 3 random variables.

The individual weights are independent since the weight of any one component part does not influence the weight of the other two.

$\therefore$ The weight of the assembly follow a normal distribution.

mean weight of the assembly

$$= 312.8 + 53.2 + 17.5 = 383.5 \text{ grams}$$

$$\text{Variance} = 2.67 + 0.85 + 0.21 = 3.73 \text{ gm}^2$$

$$\text{S.D} = \sqrt{\text{Variance}} = \sqrt{3.73} = 1.93 \text{ gms}$$

2. The four independent normal r.v X_1, X_2, X_3 and X_4 have the following means and variances.

	Mean	Variance
X_1	12	4
X_2	-5	2
X_3	8	5
X_4	10	1

Find mean and variance of

$$Q = X_1 - 2X_2 + 3X_3 - 4X_4 + 5. \text{ Find also}$$

the S.D of Q.

Soln:-

$$\begin{aligned} \text{Mean} &= E(Q) = 1E(X_1) - 2E(X_2) + 3E(X_3) - 4E(X_4) + 5 \\ &= 1 \times 12 - 2 \times (-5) + 3 \times 8 - 4(10) + 5 \\ &= 11 \end{aligned}$$

$$\begin{aligned} \text{Variance } V(Q) &= 1^2 V(X_1) + (-2)^2 V(X_2) + 3^2 V(X_3) \\ &\quad + (-4)^2 V(X_4) \\ &= 1(4) + 4(2) + 9(5) + 16(1) \\ &= 73 \end{aligned}$$

$$\text{S.D} = \sqrt{73} = 8.544$$

